NARENDER RAJU BONGONI

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PYTHON • TECHNICAL LEAD • SYSTEM ARCHITECT

Python Specialist, Solution Architect, and Engineering Leader with **20+ years** of experience designing, developing, and delivering scalable, cloud-native enterprise solutions across government, humanitarian, finance, telecom, and AI-driven platforms. Extensive hands-on expertise in **Python, Django, Django REST Framework, Flask, and FastAPI**, React **JS** specialising in solution architecture, backend engineering, RESTful API development, microservices, Role-Based Access Control (RBAC), authentication and authorisation, cloud-native applications, and enterprise system integration. **Active UK Security Clearance (SC).**

Proven track record leading architecture and engineering teams, currently s**VP of Engineering** on the founding team at **Sherpal**, where I led product architecture, engineering strategy, and the delivery of AI-powered learning platforms. Strong expertise in **AWS, Azure, Docker, Kubernetes, CI/CD, DevOps, Infrastructure as Code, PostgreSQL, Redis, and event-driven architectures**, with recent experience integrating **LLMs, Generative AI, LangChain, LangGraph, RAG, OpenAI, and Azure OpenAI** into secure, scalable enterprise

CORE TECHNICAL SKILLS

Python Frameworks: Django, Flask, FastAPI, Django REST Framework, Django CMS, Frappe
Languages: Python 3, C, C++, Perl, JavaScript, React JS , PHP
Cloud & Infra: AWS (IAM, ECS, ECR, Lambda), Azure (OpenAI, AI Search, Functions, Blob Storage), Docker, Kubernetes, Terraform, CDK
Databases: PostgreSQL, MySQL, MongoDB, Cassandra, SQLite3, Oracle, InfluxDB
AI / LLM: OpenAI APIs, Azure OpenAI, LangChain, LangGraph, Hugging Face Transformers, RAG, Prompt Engineering
CI/CD & DevOps: Jenkins, GitHub Actions, Bitbucket Pipelines, CircleCI, Bamboo, Ansible
Messaging & Search: Celery, Redis, RabbitMQ, Elasticsearch/OpenSearch, Kibana
Practices: Agile, Scrum (Certified Scrum Master), TDD, XP, Continuous Integration

EDUCATION & CERTIFICATIONS

Post Graduate Program, Artificial Intelligence & Machine Learning (Business Applications) — University of Texas at Austin
Master's in Software Engineering — Birla Institute of Technology and Science, Pilani, India
Bachelor's in Computer Science Engineering — Jawaharlal Nehru Institute of Technology, Hyderabad
Certified Professional Scrum Master I — Scrum.org | Django CMS (Open Source): Technical Committee Member & Contributor

PROFESSIONAL EXPERIENCE

Founding Engineer, VP of Engineering, Founding Team | Sthirah Inc. (Sherpal) *Mar 2025 – Present*

Python, Django, Django REST Framework, FastAPI, React JS, AWS, OpenAI, LangChain, LangGraph, React, PostgreSQL, Redis, Docker, Stripe

- Led the engineering organisation as a founding team member, defining the technical vision, product architecture, engineering roadmap, and managing cross-functional teams across Backend, Frontend, AI/ML, DevOps, QA, and Product Engineering.
- Designed, developed, and maintained scalable backend platforms using **Python, Django, Django REST Framework, and FastAPI**, delivering secure **RESTful APIs**, **Role-Based Access Control (RBAC)**, authentication and authorisation, third-party integrations, and cloud-native microservices for enterprise **web applications**.
- Architected and delivered an enterprise AI platform leveraging **OpenAI, LangChain, LangGraph, RAG, semantic search, AI agents, and personalised knowledge graphs**, enabling intelligent learning recommendations, AI-powered content workflows, and adaptive study planning.
- • Designed and implemented **AWS Lambda-based serverless workflows** using **Python, Amazon SES email templates, and Amazon EventBridge scheduled rules (cron jobs)** to automate transactional email notifications, student engagement reminders, scheduled background tasks, and recurring business processes, improving operational efficiency and reducing manual intervention.
- Led the end-to-end development of Sherpal's Phase 2 platform using **React, PostgreSQL, Redis, Docker, and AWS**, introducing personalised learning journeys, consistency analytics, performance dashboards, and scalable cloud-native architecture.
- Improved platform scalability, reliability, and performance through **Redis caching, asynchronous processing, database optimisation, API tuning, CI/CD pipelines, Docker, GitHub Actions, monitoring, and AWS infrastructure**.
- Drove engineering excellence through technical recruitment, mentoring, architecture governance, Agile delivery, stakeholder collaboration, and the implementation of **Stripe-based subscription management, payment processing, voucher redemption, tax calculation, automated billing, engineering KPIs, and AI performance metrics**.

Machine Learning Developer (Contract) | Ministry of Defence, PA Consulting *Oct 2025 – Mar 2026*

Python, Flask, FastAPI, Azure OpenAI, RAG, Multi-Agent Systems, Docker, OpenShift, Azure, REST APIs

- Built and deployed **AI-native and agentic applications using Python, Flask and FastAPI**, integrating Azure OpenAI for document automation, intelligent moderation and computer-vision annotation workflows.
- Designed and developed secure, scalable **RESTful APIs and AI microservices using Flask and FastAPI**, connecting LLM capabilities with enterprise applications, data sources and operational workflows.
- Developed **LLM and RAG pipelines** incorporating prompt engineering, enterprise knowledge retrieval, structured outputs, validation and grounded response generation.
- Designed **multi-agent and deterministic AI workflows** with controlled tool execution, workflow state, validation and **human-in-the-loop (HITL)** controls.
- Applied production AI engineering practices including **LLM evaluation, prompt and model versioning, guardrails, regression testing, observability and monitoring**.
- Containerised Python AI services using **Docker** and deployed them across **Azure and OpenShift**, supporting scalable and resilient production environments.

Senior Python Developer (Contract) | Ministry of Housing, Communities & Local Government (MHCLG)

Apr 2024 – Mar 2025

Python 3, Django, Django REST Framework, Flask, Jinja2, GitHub, AWS, Redis, PostgreSQL, GOV.UK Design System

- Designed and developed scalable **full-stack web applications using Python, Flask and ReactJS**, supporting government funding services and business workflows.
- Built secure, high-performance **RESTful APIs using Python and Flask**, providing backend services for ReactJS applications and integrations with external systems.
- Developed reusable **ReactJS components, forms and user interfaces**, integrating frontend applications with Flask REST APIs for dynamic data retrieval and submission.
- Implemented backend business logic, data validation, authentication and authorisation, exception handling and API integration using **Python and Flask**.
- Integrated Flask services with **PostgreSQL, Redis, AWS services and third-party APIs**, ensuring secure and reliable data exchange.
- Improved application performance through **PostgreSQL query optimisation, Redis caching, API optimisation and efficient frontend-backend communication**.
- Implemented automated testing, logging, error handling and CI/CD practices to support reliable production releases.
- Collaborated with product, QA, DevOps and engineering teams on **solution design, API contracts, code reviews and production delivery**.

Tech Lead (Contract) | Waracle / Ciena

Sep 2022 – Feb 2024

Python 3, Django, Django REST Framework, Keycloak, IAM, OAuth 2.0, OpenID Connect (OIDC), SSO, RBAC, Docker, Kubernetes, AWS, Azure, YANG, SD-WAN, KubeMQ, ReactJS

- Contributed to the **assessment and design of the Identity and Access Management (IAM) architecture** for Ciena's network-management and portal applications, reviewing authentication, authorisation, user-access and API-security requirements.
- Evaluated application and microservice authentication requirements and helped define a **centralised IAM approach using Keycloak**, providing a consistent identity and access-control layer across web applications and backend services.
- Designed and implemented **Keycloak integration** for Python/Django applications and REST APIs, supporting Single Sign-On (SSO), authentication, authorisation and Role-Based Access Control (RBAC).
- Implemented **OAuth 2.0 and OpenID Connect (OIDC)** based authentication and token-based API security, including Keycloak clients, roles, groups, role mappings and protected service endpoints.
- Worked with application and platform teams to define **identity flows, access-control boundaries and service-to-service security**, ensuring authentication and authorisation were consistently enforced across distributed services.
- Reviewed IAM and application-security requirements as part of technical design discussions, identifying integration risks and recommending appropriate **Keycloak configuration and security patterns**.
- Designed and developed scalable backend applications and RESTful APIs using **Python, Django and Django REST Framework** for network-device configuration, SD-WAN domain/template management and enterprise portal applications.
- Built and enhanced containerised microservices using **Docker and Kubernetes** across AWS and Azure, incorporating secure IAM integration into the wider cloud-native architecture.

- Developed comprehensive TDD unit and integration tests and contributed to architecture reviews, code reviews, CI/CD and production releases.

Python Django API Developer (Contract) | QA.com *Dec 2020 – Sep 2022*

Python 3, Django, Django REST Framework, FastAPI, React JS, Docker, Celery, RabbitMQ, PostgreSQL, AWS, AWS Lambda, Jenkins, Bitbucket, Stripe

- Designed and developed scalable portal applications using **Python, Django, Django REST Framework and ReactJS**, delivering secure RESTful APIs for checkout, organisation management and enterprise integrations.
- Implemented **OAuth 2.0-based authentication and authorisation** for web applications and REST APIs, securing access to backend services and protected application resources.
- Designed and implemented **Role-Based Access Control (RBAC)**, authentication and authorisation across portal applications and APIs, ensuring users and services could access only permitted resources and operations.
- Integrated authentication and access-control mechanisms with Django REST APIs and ReactJS applications, supporting secure **token-based API communication** between frontend and backend services.
- Built cloud-native backend services and microservices using **FastAPI, Celery, RabbitMQ, Docker and AWS**, including Stripe payment APIs, AWS Lambda webhook processing and notification services.
- Implemented comprehensive **TDD/BDD, unit, integration, API and mock-database testing**, contributing to Jenkins and Bitbucket CI/CD pipelines for secure, reliable production releases.

Python/Django Senior Developer | Nyman (Eliga Services) / Fidelity International

Oct 2017- Jan 2020 & July2020-Dec 2020

Python 3, Django, Django CMS, Flask, Docker, Elasticsearch, Celery, RabbitMQ, PostgreSQL, AWS, AWS Lambda, Terraform, Jenkins, Bitbucket, Pandas

- Designed and developed enterprise web applications and reusable **Django CMS** components, RESTful APIs, article management modules, webhooks, and Elasticsearch-powered global search, leading the successful migration of **40+ Fidelity International websites** across Europe and APAC to a multi-site Django CMS platform.
- Developed scalable backend services using **Python, Django, Flask, Celery, RabbitMQ, PostgreSQL, and AWS**, implementing content migration frameworks, Django CMS 4 migrations, AWS infrastructure automation with Terraform, CI/CD pipelines, monitoring with New Relic, and cloud-native deployment practices.
- Led technical delivery across distributed teams in the UK, India, Dalian, and Tokyo, performing architecture reviews, peer code reviews, mentoring developers, and implementing comprehensive **TDD** using Django Unit Tests and Factory Boy to ensure high-quality, production-ready releases.

Python/Django Senior Developer (Contract)| Willis Tower Watson

Feb 2017- Oct' 2017

Python 3, Django, Flask, Docker, Elasticsearch, Celery, RabbitMQ, PostgreSQL, AWS, AWS Lambda, Terraform, Jenkins, Bitbucket.

- Development RESTful Services for regulatory reports,
- Development of Python Backend and Django Restful API's pension Administration application modules
- Write TDD unit tests
- Write BDD behave tests for automating using selenium

EARLIER CAREER (2003 – 2017)

Progressive Python/Django and backend engineering roles delivering CMS migrations, search infrastructure, and enterprise systems, including:

- Willis Towers Watson, IBM, BSKYB, CapriceCloud, Amadeus Services (2012–2017) — Python/Django backend and REST API development for pensions administration, storage automation, search migration, and enterprise WebUI/CLI systems.
- EMC², Quantum, Rocsys Technologies, Vedams Software Solutions, DLRL (2003–2012) — Python, C/C++ engineering for network-attached storage automation, replication, firewall, and radar management systems.